Participation Explains Half the Gender Gap at the Top of Chess

A test on the FIDE rating list of November 2026

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Abstract

Since men hold almost nine of every ten chess ratings, Bilalić, Smallbone, McLeod and Gobet (2009) argued that the best men outrate the best women mostly because there are more of them, and in German data they found that numbers alone accounted for 96 percent of the gap between the 100 best men and the 100 best women. To repeat their test, we take the 223,908 active players on the FIDE standard list of November 2026 and compute the gap expected from participation by randomly permuting the sex labels while holding the numbers of men and women fixed. About half of the gap is explained, with participation accounting for 140 of the 268 rating points that separate the world's top 100 pairs and for 56 percent of the German gap. Looking only at the best man and the best woman, the share falls to 24 percent. Even below the top the two distributions differ, since women's mean rating is 115 points lower than men's, and 66 points lower among players born after 2005. Most surprisingly, across 55 federations with at least 100 active women, participation explains less of the top gap where women are a larger share of players. A count of players thus takes us halfway. The other half could come from differences in training, age at entry, games played, opponents in women's events or the truncation of the list at a rating of 1,400, and this test cannot tell them apart. Explaining that half is a task for other data.

1 The test

Nowhere is the scarcity of women at the top more often cited as evidence of sex differences in ability than in chess. Earlier work by Bilalić et al. (2009) argued instead that the scarcity at the top follows mostly from the scarcity of women at every level. If men and women draw ratings from the same distribution, the best of 100,000 men will be stronger than the best of 10,000 women, because the extreme of a larger sample is higher. They compared the i-th best man and the i-th best woman in Germany for $i = 1, \ldots, 100$ and found that the gap expected from participation alone accounted for 96 percent of the actual gap. However, Knapp (2010) argued that participation alone does not account for the whole gap.

Each of the 223,908 players on the FIDE standard rating list of 12 November 2026 who have a standard rating and are not flagged inactive enters our test. Rather than assume a distribution for the tail, we compute the expected gap by permutation. Specifically, within each federation, or the world, we randomly reassign the sex labels across players, keeping the numbers of men and women fixed, and compute the same statistic, averaging over 100 to 500 permutations. In this way the expected gap is the gap that participation alone produces if the two sexes share the observed pooled distribution, and the share explained is the expected gap divided by the actual gap. Differences are measured in rating points throughout.

	Actual gap	Expected from participation	Share explained	Women
World, best man vs best woman	227	56	24%	24,150
World, top 10 averages	234	92	39%	
World, top 100 pairs	268	140	52%	
Germany, top 100 pairs, active players	378	213	56%	1,157
including inactive players	305	173	57%	2,724
players born 2006 or later	389	226	58%	511
55 federations, top 20 pairs, pooled			64%	
median federation			62%	

Table 1: Rating gap between men and women at the top of the FIDE standard list, 12 November 2026, in rating points. Top-K pairs is the mean of the differences between the i-th best man and the i-th best woman, $i = 1, \ldots, K$. The expected gap is the mean of the same statistic over random permutations of the sex labels within the group, holding the number of each sex fixed. Active players only unless stated. The federation rows cover federations with at least 100 active rated women.

2 Results

Even at the top, participation explains only about half of the gap (Table 1). Comparing the world's 100 best men with its 100 best women, the average gap is 268 points, and random sex labels with the same numbers of each sex produce a gap of 140, so 52 percent is explained. At the very top the share is lower, since the best man is rated 227 points above the best woman and participation accounts for 56 of them. Nor does Germany, the setting of the original study, look different, with 56 percent explained for the top 100 pairs. Changing the sample hardly matters, as the share is 57 percent when inactive players are included and 58 percent for players born in 2006 or later. Looking across the 55 federations with at least 100 active women, participation explains 64 percent of the gap between the 20 best men and the 20 best women, pooling across federations.

A pure participation account also predicts no gap away from the top, and there is one. In the whole list women's mean rating is 1,630 and men's 1,745. Measured within birth cohorts the gap is smaller but persistent, at 112 points for players born before 1970, 68 for those born in 1970-89, 64 for 1990-2005 and 66 for players born after 2005, a cohort in which women are 18 percent of rated players. The gap in mean ratings is smaller where women are a larger share of players, with a correlation of -0.49 across federations. Higher female participation, however, goes with a smaller share of the top gap explained by participation, with a correlation of -0.59. Examples are Russia, where women are 21 percent of active players and participation explains 35 percent of the top-20 gap, and the Netherlands, Sweden and Belgium, where women are 4 percent and participation explains 74 to 82 percent.

3 Scope

FIDE lists only players with an international rating, and the lowest rating on the list is 1,400, the floor FIDE introduced in 2024. Unrated and weaker players are therefore missing, and since women are overrepresented in the lower part of the list they are probably overrepresented below the floor as well, so the participation ratios describe internationally rated players and not all players. Like any permutation test, ours treats the pooled distribution as common to both sexes, so any mean difference appears in the unexplained part. Leading candidates for that difference include ability, training, age at entry, the number of games played and the strength of opponents in women's events, and the test cannot separate them. Part of the contrast with the 2009 study may also come from the data, since Bilalić et al. used the German national rating list, which includes many players without a FIDE rating.

On these numbers participation is half of the story, and the other half is open. In chess terms the position is level. No further move is available to this test. The first letter of each sentence in the abstract and the text, read in order, gives our verdict.

References

[1] Bilalić, M., K. Smallbone, P. McLeod and F. Gobet (2009). Why are (the best) women so good at chess? Participation rates and gender differences in intellectual domains. *Proceedings of the Royal Society B* 276, 1161–1165.

[2] Knapp, M. (2010). Are participation rates sufficient to explain gender differences in chess performance? *Proceedings of the Royal Society B* 277, 2269–2270.